CHAPTER 2

# Quadratics: forms and choice of method

**Chapter path**

**Objectives.** Choose a form; solve an equation or inequality; read an extremum; prove the formula for the roots.
**Prerequisites.** Special products, zero-product property, sign of a product, square roots.
**Yearly reference point.** 22 September·1 October: 8 hours

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Expand ( $x-3)^2$.
2. Factorise $x^2-16$.
3. Solve ( $x-2)(x+5)=0$.
4. Solve $x^2$ = 9 in $\mathbb{R}$.
5. Give the sign of 2 $x-6$.
6. Determine the minimum of ( $x-1)^2+4$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** Present two methods for factorising a quadratic. In the proof by discriminant, name the three cases and explain why they are exhaustive.

## Investigate before formalising

A rectangle has an area of 24 m $^2$ and a perimeter of 20 m. Without randomly trying all the dimensions, determine its sides. Start with a drawing and explain the two constraints.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

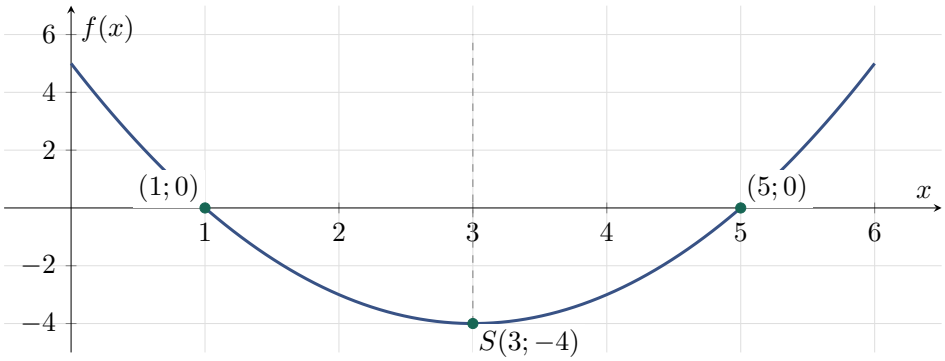
## 2.1 Three forms, three readings

**Definition and results**

A quadratic function can be written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The expanded form facilitates the calculation of $f(0)$ ; the vertex form $a(x-\alpha)^2 + \beta$ reveals vertex and extremum; a factorised form $a(x - r_1)(x - r_2)$ reveals the roots and the sign. The roots are the solutions of $f(x) = 0$.

**Status and justification.** The completion of the square is based on $x^2+2ux = (x+u)^2-u^2$. The existence of the canonical form is proved by expanding; computing it in general is not an automatic skill required.

**Worked example.** $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is $-4$ at 3 ; the roots are 1 and 5; the y-intercept is 5.




Curve of $f(x) = (x-3)^2 - 4$. The two roots and the vertex are marked; the axis of symmetry is the right $x = 3$.

**Watch out**

A factorised expression is not obtained by putting each term separately in parentheses.

**Immediate application.** Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and its minimum.

## 2.2 Discriminant and solving

**Definition and results**

For $ax^2 + bx + c$ = 0 with $a \neq 0$, set ?= $b^2 - 4ac$. If $< 0$, no real root; if ?= 0, one root $-\frac{b}{2a}$ ; if $> 0$, two roots $\frac{-b - \sqrt{\Delta}}{2a}$ and $\frac{-b + \sqrt{\Delta}}{2a}$. To order roots, take the sign of a into account.

**Status and justification.** Multiplying the equation by 4 $a \neq 0$, we obtain (2 $ax + b)^2$ = ?. A square is positive or zero: the three cases follow. If ? $> 0$, $2ax + b = \pm \sqrt{\Delta}$. Splitting into cases avoids take the square root of a negative number.

**Worked example.** $2x^2 + x - 3 = 0$ : $\Delta = 1 + 24 = 25$ ; $x = \frac{-1 \pm 5}{4}$i.e. 1 and $-\frac{3}{2}$. Check: $(x - 1)(2x + 3) = 2x^2 + x - 3$.

**Watch out**

The denominator is 2 $a$, not 2 ; the sign of b is taken with its brackets in $b^2$.

**Immediate application.** Solve 3 $x^2 - 6x$ + 3 = 0 and $x^2 + 2x + 5 = 0$.

## 2.3 Factorising without automatically using the discriminant

**Definition and results**

If two real roots exist, their sum is $-\frac{b}{a}$ and their product $\frac{c}{a}$. An apparent root or a special product often gives a shorter factorisation. Polynomials of degree 2 vanishing at two distinct real numbers r and s are exactly $a(x - r)(x - s)$ with $a \neq 0$.

**Status and justification.** Expanding $a(x - r)(x - s) = ax^2 - a(r + s)x + ars$, identifying of the coefficients gives the relations. Conversely, if r and s have this sum and this product, the expanding recovers the polynomial.

**Worked example.** $x^2 - 7x$ + 12 : 3 and 4 have sum 7 and product 12. Thus ( $x - 3)(x - 4$. For additionally impose $f(0)$ = 24, we need 12 $a$ = 24, so $a = 2$.

**Watch out**

The sum of the roots is $-\frac{b}{a}$ ; do not forget the minus sign. A single root is not enough to determine a.

**Immediate application.** Determine the quadratic polynomial f such that $f(1) = f(4)$ = 0 and $f(0) = -8$.

## 2.4 Sign, inequalities and optimisation

**Definition and results**

For two distinct roots $r < s$, the quadratic has the sign of a outside [ $r; s$] and the sign opposite to a between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it has the sign of a away from it. Optimisation is done on the domain of the problem, which may limit access to the vertex.

**Status and justification.** The sign table of the factors $a(x - r)(x - s)$ gives the result. The canonical form compares $f(x)$ à $\beta$ using the sign of $a(x - \alpha)^2$.

**Worked example.** $-2(x - 1)(x - 5) \geq 0$ is equivalent to $x \in [1; 5]$. For $A(x) = x(12 - x)$ on [0; 12], $A(x) = 36 - (x - 6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has area maximum of 36 m $^2$ when it is square.

**Watch out**

In a strict inequality, the roots are excluded. Check the geometric bounds before conclude.

**Immediate application.** Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on [0; 3].

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

### Exercise 2.1 · Complete a square

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in the form ( $x - 4)^2 + k$, where $k$ is a real number to be determined.
2. Deduce a factorisation of $f(x)$ using a remarkable identity.
3. Draw the sign table of $f$, then solve in $\mathbb{R}$ Inequation $f(x) \leq 0$.

### Exercise 2.2 · Discussing without heavy calculation

We study the functions $f(x) = x^2 + 4$, $g(x) = (x + 2)^2$ and $h(x) = -3(x - 1)(x + 4)$, all defined on $\mathbb{R}$.

1. For each function, determine any real roots by working directly from the form given.
2. State the sign of each function on $\mathbb{R}$. State explicitly the values at which it vanishes.
3. Explain why computing a discriminant is unnecessary in all three cases.

## Exercise 2.3 · Selecting a Strategy

We want to solve in $\mathbb{R}$ the following three equations:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: special product, finding two numbers with known sum and product, or discriminant. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Check the solutions of the first two equations by substitution. For the third, check the sum and product of the roots obtained.

## Exercise 2.4 · Find a parabola

The curve of a polynomial function $f$ of degree 2 cuts the x-axis at the points with x-coordinates $-2$ and 3. It also passes through the point $A(1; 12)$.

1. Determine an expression of $f(x)$ from these three pieces of information. State the chosen form and why it is suitable.
2. Expand the expression found. Deduce the y-intercept of the curve.
3. Check that the function obtained satisfies the three initial pieces of information.

## Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of fencing to enclose three sides of a rectangular vegetable garden. The fourth side runs along a wall and needs no fencing. Let $x$ the length, in metres, of each of the two sides perpendicular to the wall. The whole length of fencing is used.

1. Draw a diagram and express the length of the side parallel to the wall in terms of $x$. State the values of $x$ for which the rectangle is not degenerate.
2. Express the area $A(x)$, in square meters. Determine its maximum value and the corresponding dimensions, by choosing the method of your choice.
3. Check that these dimensions use exactly 20 m of fencing.

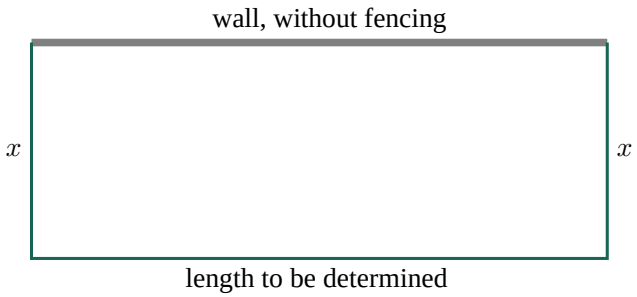



Diagram of the vegetable garden: only the three green lines use the 20 m of fencing. The proportions of the drawing does not determine the optimal dimensions.

### Exercise 2.6 · A Parameter

For every real number $m$, consider the equation with real unknown $x$ :

$$x^2 - 2x + m = 0.$$

The number $m$ is a parameter: it remains fixed when the equation is solved.

1. Determine the values of $m$ for which the equation has two distinct real solutions.
2. Determine the values of $m$ for which it has a unique real solution, then those for which it has none.
3. Give the solution when it is unique. Explain why the three cases considered cover all values of $m$.

### Exercise 2.7 · Link with sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for all natural $n$. We seek all indices at which the term of the sequence is strictly negative.

1. First solve in $\mathbb{R}$ Inequation $x^2 - 5x + 4 < 0$.
2. Deduce all the natural numbers $n$ such that $u_n < 0$. Justify why a real interval does not is not the final answer here.

### Exercise 2.8 · Delayed review

A student studies the function $f$ defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She writes: « The minimum value $-5$ and it is attained at $x = 3$. »

1. Say what is correct and what is wrong in this sentence. Rewrite it correctly in justifying your answer.
2. Solve in $\mathbb{R}$ Inequation $f(x) \leq -3$, then check the two bounds of the obtained set.

**Independent work.** Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, exercise 43 p. 99, questions a and c.** First find an obvious root, then the other using the product of the roots.

**CQFD, exercise 49 p. 99, questions a and b.** Express the sum and the product as an equation ; do not assume in advance that the numbers exist.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. The roots of $(x+1)(x-4)$ are

   A. 1 and 4

   B. $-1$ and $-4$

   C. $-1$ and 4

   D. 1 and $-4$

2. The discriminant of $x^2 - 2x + 3$ is

   A. $-8$

   B. $8$

   C. $16$

   D. $-2$

3. The vertex of $(x-2)^2 + 5$ is

   A. $(2;-5)$

   B. $(-2;5)$

   C. $(5;2)$

   D. $(2;5)$

4. The sum of the roots of $2x^2 - 6x + 1$ is

   A. $6$

   B. $3$

   C. $-3$

   D. $\dfrac{1}{2}$

5. $x^2 - 4 < 0$ is equivalent to

   A. $x > 2$

   B. $x < -2$

   C. $x \in ]-2;2[$

   D. $x \in [-2;2]$

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $x^2 + 6x + 9$ factorises as

   A. $(x-3)^2$

   B. $(x+9)(x+1)$

   C. $(x+6)^2$

   D. $(x+3)^2$

2. The product of the roots of $3x^2 + x - 2$ is

   A. $-\dfrac{2}{3}$

B. $-\dfrac{1}{3}$

C. $\dfrac{2}{3}$

D. $-2$

3. $-x^2 + 4x$ reaches its maximum on [0; 1] at

A. 2

B. 0

C. 1

D. 4

4. A double root requires

A. $a = 0$

B. $\Delta = 0$

C. $b = 0$

D. $c = 0$

5. The quadratic polynomials with roots 0 and 2 are

A. $a(x + 2)$

B. $a(x - 2)^2$

C. $x(x - 2)$ only

D. $ax(x - 2)$, $a \neq 0$

## True or false? Justify.

1. A quadratic function always has two real roots.

2. A quadratic with two distinct roots is negative between them.

3. For every real x, ( $x - 2)^2 \geq 0$.

**Personal review**

I justify the choice of form; I distinguish roots and vertex; I finish an inequality with a
together; I respect the domain of the model.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and
a mistake to avoid.